

LABORATORY OBSERVATION

Course: Full Stack Web Development

Course Code: 23CM4121

Experiment No.: 2

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1. TITLE

Design a Webpage Using HTML5 and CSS3

2. AIM

To design a webpage using HTML5 semantic elements and to style it using CSS3, in order to understand how structure and presentation work together on the web.

3. OBJECTIVE

The objectives of this experiment are:

- To understand HTML5 semantic tags such as header, nav, main, section, article and footer.
- To apply CSS3 styling using selectors, the box model, colors and fonts.
- To link an external CSS file to an HTML page.
- To create a visually styled webpage combining HTML5 structure and CSS3 styling.

4. THEORY

HTML5 introduced semantic elements such as <header>, <nav>, <main>, <section>, <article> and <footer>, which describe the meaning of the content they contain rather than just its appearance. This makes web pages easier to read, maintain and access.

CSS3 (Cascading Style Sheets Level 3) is used to control the visual presentation of HTML content. It allows selectors (tag, class, id), the box model (margin, border, padding), colors, fonts, backgrounds and layout properties to be defined.

CSS can be applied to HTML in three ways: inline (using the style attribute), internal (using a <style> block in the <head>), or external (using a separate .css file linked with the <link> tag). The external method is preferred as it keeps structure and style separate.

5. ALGORITHM / PROCEDURE

1. Create a new HTML file and save it with a .html extension.
2. Add semantic HTML5 elements: <header>, <nav>, <main>, <section> and <footer>.
3. Add appropriate content inside each section (heading, text, links).

4. Create a separate CSS file named style.css.
5. Define selectors for the various HTML elements using tag, class and id selectors.
6. Apply styles such as background-color, color, font-family, padding, margin and border-radius.
7. Link the CSS file to the HTML page using the <link> tag inside <head>.
8. Save both files in the same folder.
9. Open the HTML file in a web browser to view the styled webpage.

6. PROGRAM

index.html

```
<!DOCTYPE html>
<html>
<head>
  <title>HTML5 & CSS3 Webpage</title>
  <link rel="stylesheet" href="style.css">
</head>
<body>

  <header>
    <h1>My Portfolio</h1>
    <nav>
      <a href="#about">About</a>
      <a href="#project">Project</a>
      <a href="#contact">Contact</a>
    </nav>
  </header>

  <main>
    <section id="about">
      <h2>About Me</h2>
      <p>I am a student learning full stack web development.</p>
    </section>

    <article id="project">
      <h3>My Project</h3>
      <p>This webpage demonstrates HTML5 semantic tags styled using CSS3.</p>
    </article>
  </main>

  <footer id="contact">
    <p>&copy; 2026 Gorapalli Sravya</p>
  </footer>

</body>
</html>
```

style.css

```
body {
  font-family: Arial, sans-serif;
  margin: 0;
  background-color: #f4f4f4;
}

header {
  background-color: #2c3e50;
```

```

    color: white;
    padding: 20px;
    text-align: center;
}

nav a {
    color: white;
    margin: 0 10px;
    text-decoration: none;
    font-weight: bold;
}

section, article {
    background-color: white;
    margin: 20px;
    padding: 15px;
    border-radius: 8px;
    box-shadow: 0 0 5px rgba(0,0,0,0.1);
}

footer {
    text-align: center;
    background-color: #2c3e50;
    color: white;
    padding: 10px;
}

```

7. CODE EXPLANATION

Tag / Code	Explanation
<header>	Defines the top section of the page (title, navigation)
<nav>	Defines a section of navigation links
<main>	Defines the main content area of the page
<section>	Defines a thematic grouping of content
<article>	Defines independent, self-contained content
<footer>	Defines the bottom section of the page
<link rel="stylesheet">	Links an external CSS file to the HTML page
font-family / color	CSS properties that set the text font and color
background-color	CSS property that sets the background color
padding / margin	CSS box-model properties that control spacing
border-radius	CSS property that rounds element corners
box-shadow	CSS property that adds a shadow effect to an element

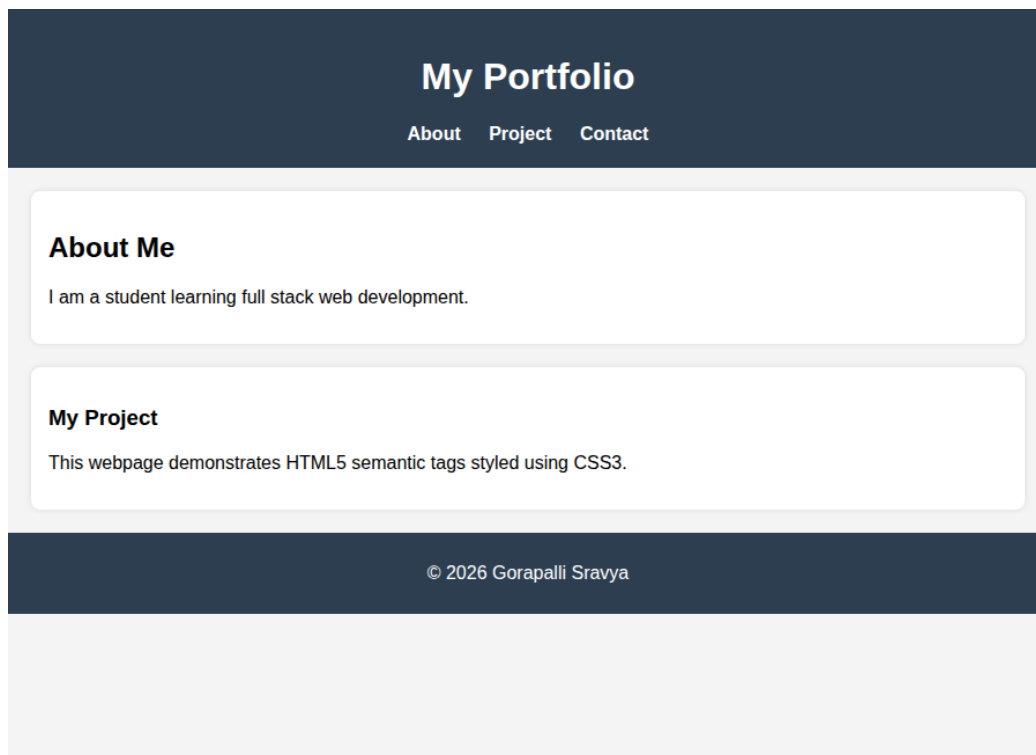
8. TEST CASES AND EXECUTION DATA

The HTML file was opened in a browser with the CSS file linked, and each styled section was verified.

Test Case	Description	Expected Result	Actual Result	Status
TC-01	Open index.html in a browser	Page loads with styled header, nav and sections	Page loaded with styling applied	Pass
TC-02	Check CSS is applied	Background colors and fonts render as styled	Colors and fonts rendered as styled	Pass
TC-03	Click navigation links	Links scroll to the respective sections	Links navigated correctly	Pass
TC-04	Resize the browser window	Layout remains readable	Layout remained readable	Pass
TC-05	Check footer	Footer displays at the bottom with copyright text	Footer displayed correctly	Pass

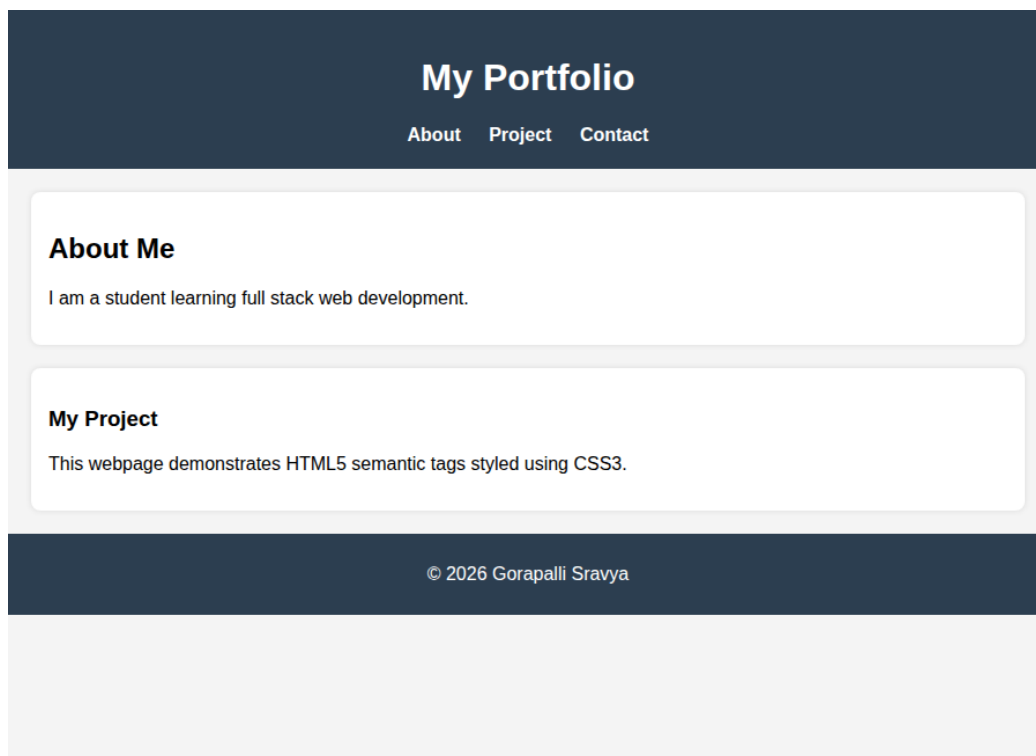
Test Case Output

TC-01 — *Page loads with styled header, nav and sections*



Page loaded with header, navigation and styled sections

TC-02 — *Background colors and fonts render as styled*



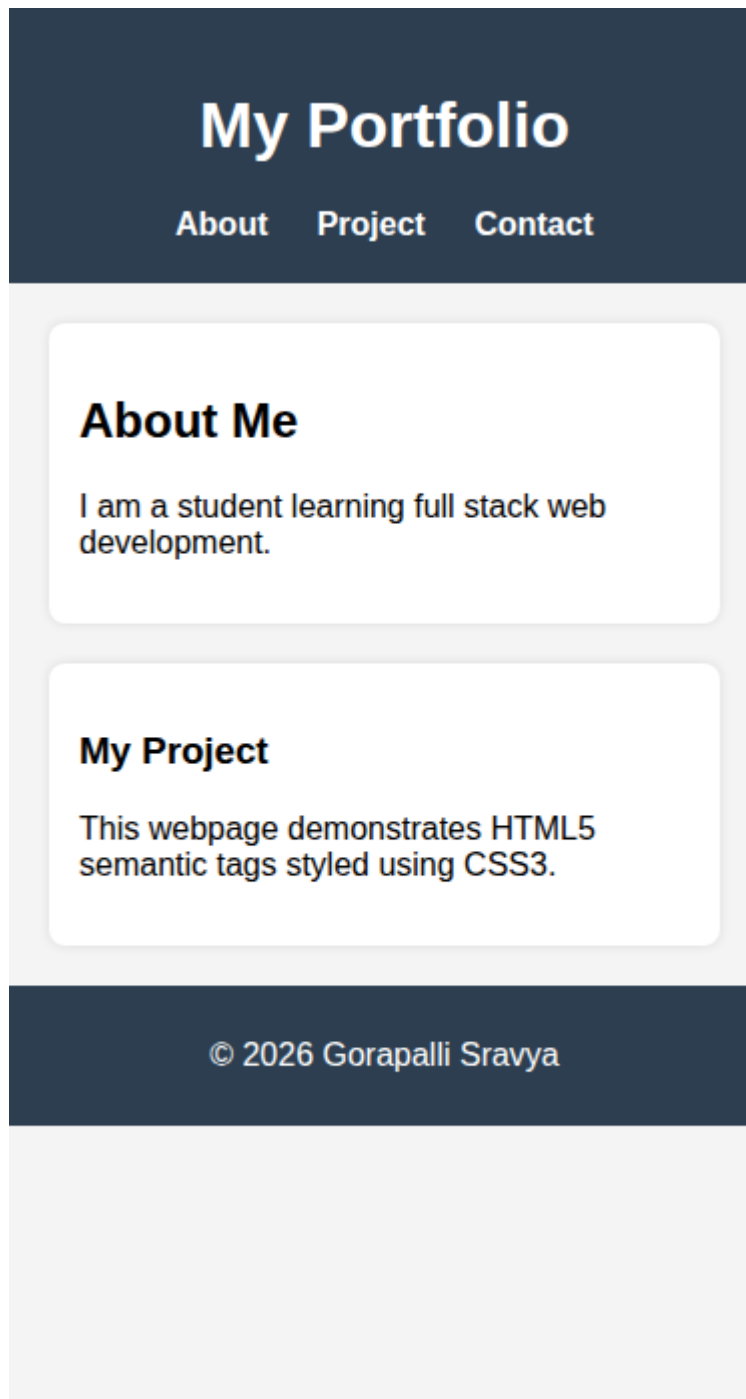
Dark header, card backgrounds and fonts rendered as styled in CSS3

TC-03 — *Links scroll to the respective sections*



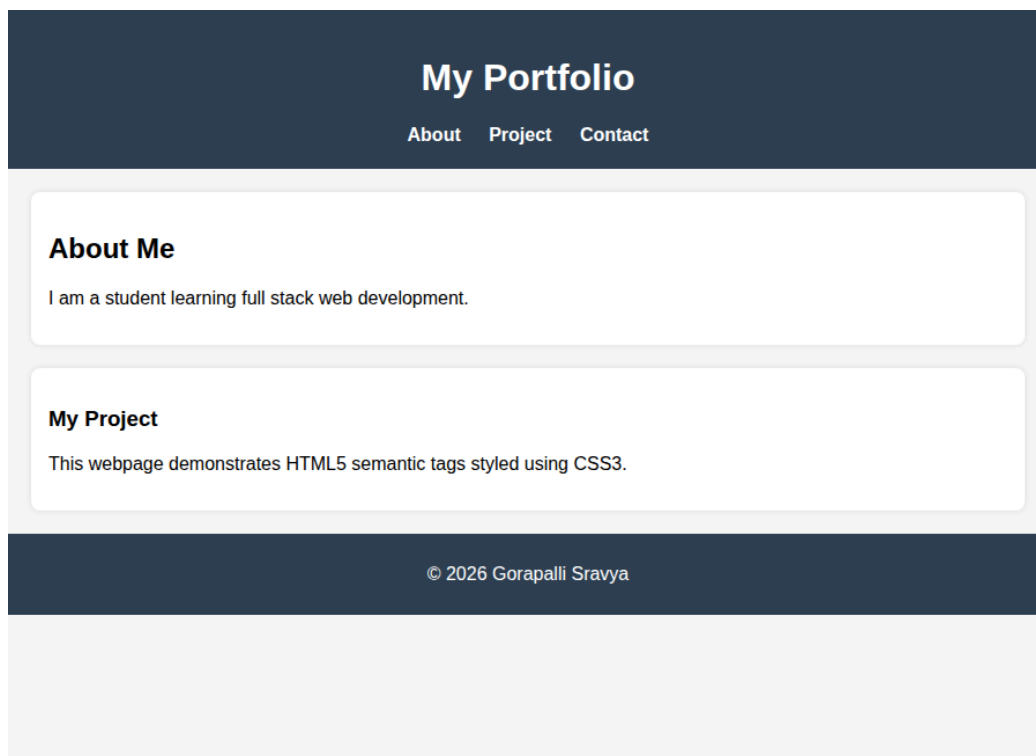
Page after clicking the 'Project' nav link

TC-04 — *Layout remains readable on resize*



Layout at a narrow (375px) viewport width — remains readable and stacked

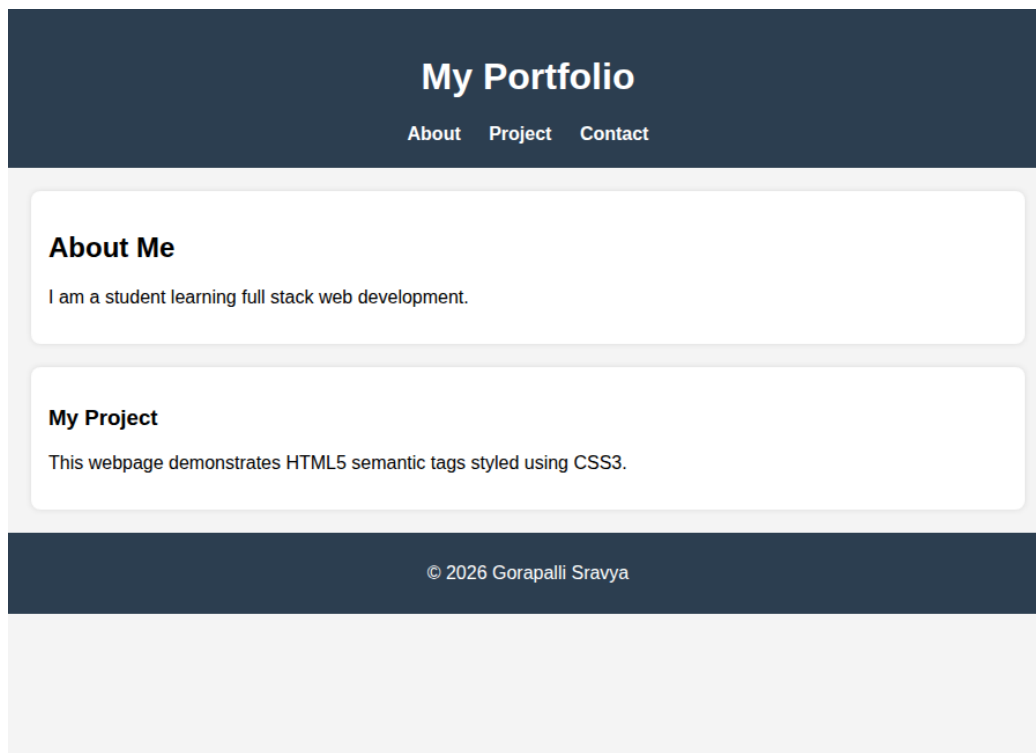
TC-05 — Footer displays at the bottom with copyright text



Footer displaying the copyright line

9. OUTPUT

Output: Styled Webpage Rendered in Browser



Webpage rendered with HTML5 semantic structure and CSS3 styling applied

10. RESULT

Thus, a webpage was successfully designed using HTML5 semantic elements and styled using CSS3, and the output was verified in a web browser.